

Chapter 13 - Measure of Dispersion

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#Review

The measures which condense the huge mass of data into a single value representing all of them and around which most of the data tend to concentrate, are known as averages.

The measure of such a single value is known as the measure of central tendency.

Arithmetic Mean

When the sum of the set of observations of the same kind is divided by the number of observations, we get a unique value known as the arithmetic mean (or simply mean).

×	Individual series	Discrete series	Continuous series
Direct method	$\bar{x} = \frac{\Sigma x}{n}$	$\bar{x} = \frac{\Sigma fx}{N}$	
Deviation method ($a \rightarrow$ assumed mean, $d = x - a$, $d' = \frac{d}{h}$)	$\bar{x} = a + \frac{\Sigma d}{n}$	$\bar{x} = a + \frac{\Sigma fd}{n}$	$\bar{x} = a + \frac{\Sigma fd'}{n} \times h$

Combined Mean

If the arithmetic means of two series having the size n_1 and n_2 are $\bar{x}_1$ and $\bar{x}_2$ then their combined mean is given by

$$\bar{x}_{12} = \frac{n_1 \bar{x}_1 + n_2 \bar{x}_2}{n_1 + n_2}$$

Partition Values

Partition values are statistical measures (specifically quartiles, deciles, and percentiles) that divide an ordered dataset into equal, actionable segments. They are used to analyze data distribution, identify outliers, and summarize large datasets.

×	Individual series	Discrete series	Continuous series
Median	$M_d = \left(\frac{n+1}{2}\right)^{th}$ term where, $n \rightarrow$ no. of observations	$M_d = \left(\frac{N+1}{2}\right)^{th}$ term where, $N \rightarrow$ total frequency	$M_d = L + \frac{\frac{N}{2} - c.f.}{f} \times h$ where, $L \rightarrow$ lower limit, $N \rightarrow$ total frequency, $c.f. \rightarrow$ cumulative frequency of preceding class $f \rightarrow$ frequency of median class $h \rightarrow$ height of class

×	Individual series	Discrete series	Continuous series
Quartile	$Q_1 = \left(\frac{n+1}{4}\right)^{th}$ term, $Q_3 = \left(\frac{3(n+1)}{4}\right)^{th}$ term, where, $n \rightarrow$ no. of observations	$Q_1 = \left(\frac{N+1}{4}\right)^{th}$ term, $Q_3 = \left(\frac{3(N+1)}{4}\right)^{th}$ term where, $N \rightarrow$ total frequency	$Q_1 = L + \frac{\frac{N}{4} - c.f.}{f} \times h,$ $Q_3 = L + \frac{\frac{3N}{4} - c.f.}{f} \times h$ where, $L \rightarrow$ lower limit, $N \rightarrow$ total frequency, $c.f. \rightarrow$ cumulative frequency of preceding class $f \rightarrow$ frequency of median class $h \rightarrow$ height of class

Mode

A variate value occurring maximum number of times, is known as mode.

$$M_o = L + \frac{\Delta_1}{\Delta_1 + \Delta_2} \times h, \text{ where } \Delta_1 = f_1 - f_0 \text{ and } \Delta_2 = f_1 - f_2$$

Relation between mean, median and mode

For symmetric distribution	For unsymmetric distribution
Mean = Median = Mode	Mode = 3 Median - 2 Mean or $M_o = 3M_d - 2\bar{x}$ This relation is known as Empirical relation.

Dispersion

Dispersion means distribution or scattering. In statistics, Dispersion is defined as the measure of scatterness or variation of the items from the central value. The measure of dispersion is of two types:

Absolute Measure of Dispersion	Relative Measure of Dispersion
These measures provide the actual amount of variation within a data set in its original units of measurement. Some of the absolute measures of dispersion are: • Range • Quartile Deviation • Mean Deviation • Standard Deviation • Variance	These are used to compare the variability of two or more datasets that may have different units or drastically different averages. They are dimensionless (unitless). Some relative measures of dispersion are: • Coefficient of Range • Coefficient of Quartile Deviation • Coefficient of Mean Deviation • Coefficient of Standard Deviation • Coefficient of Variation

#Standard Deviation

Standard deviation is defined as the positive square root of the mean of the square of the deviations taken from the arithmetic mean. It is denoted by σ .

S.D.	Individual series	Discrete series	Continuous series
Actual mean method	$\sigma = \sqrt{\frac{\Sigma(x - \bar{x})^2}{n}} = \sqrt{\frac{\Sigma x^2}{n} - \left(\frac{\Sigma x}{n}\right)^2}$ where, $n \rightarrow$ no. of observations	$\sigma = \sqrt{\frac{\Sigma f(x - \bar{x})^2}{N}} = \sqrt{\frac{\Sigma f x^2}{N} - \left(\frac{\Sigma f x}{N}\right)^2}$ where, $N \rightarrow$ total frequency	
Assumed mean method	$\sigma = \sqrt{\frac{\Sigma d^2}{n} - \left(\frac{\Sigma d}{n}\right)^2}$ where, $d = x - a$ $n \rightarrow$ no. of observations	$\sigma = \sqrt{\frac{\Sigma f d^2}{N} - \left(\frac{\Sigma f d}{N}\right)^2}$ where, $d = x - a$ $N \rightarrow$ total frequency	

S.D. by Step Deviation Method: $\sigma = h \times \sqrt{\frac{\Sigma f d'^2}{N} - \left(\frac{\Sigma f d'}{N}\right)^2}$ where $d' = \frac{x - a}{h}$ and a is assumed mean.

Merits of Standard Deviation

1. It is rigidly defined.
2. It is based on all the observations.
3. It is least affected by fluctuation of sampling.
4. It is suitable for further mathematical treatment.

Demerits of Standard Deviation

1. It is difficult to compute.
2. It gives greater weight to the extreme values and less to those which are near to the mean.
3. Standard deviation cannot be calculated for open end classes.

Combined Standard Deviation

Let n_1 and n_2 be the sizes, $\bar{x}_1$ and $\bar{x}_2$ the arithmetic means and σ_1 and σ_2 the respective standard deviation of two component series then their compound standard deviation is given by

$$\sigma_{12} = \sqrt{\frac{n_1\sigma_1^2 + n_2\sigma_2^2 + n_1d_1^2 + n_2d_2^2}{n_1 + n_2}}$$

where, $d_1 = \bar{x}_1 - \bar{x}_{12}$, $d_2 = \bar{x}_2 - \bar{x}_{12}$ and $\bar{x}_{12} = \frac{n_1\bar{x}_1 + n_2\bar{x}_2}{n_1 + n_2}$

Properties of Standard deviation

1. In a discrete distribution, the standard deviation is not less than the mean deviation from the mean.
2. Standard deviation is the least possible of root mean square deviation.
3. Standard deviation is independent of change of origin but not of scale.

#Variance

The square of the standard deviation is known as the variance. It is denoted by σ^2 and its formula is given by

$$\sigma^2 = \frac{\sum f(x - \bar{x})^2}{N} = \frac{\sum fx^2}{N} - \left(\frac{\sum fx}{N}\right)^2$$

Mean square deviation

The mean square deviation denoted by s^2 is defined by

$$s^2 = \frac{\sum f(x - a)^2}{N} \quad \text{where } a \text{ is an arbitrary number.}$$

Root mean square deviation

The positive square root of the mean square deviation is known as root mean square deviation.

It is denoted by s and its formula is given by

$$s = \sqrt{\frac{\sum f(x - a)^2}{N}}$$

Coefficient of Variation

Standard deviation is the absolute measure of dispersion. The relative measure of dispersion based on the standard deviation is known as the coefficient of standard deviation. Symbolically, it is defined by

$$\text{Coefficient of s.d.} = \frac{\sigma}{\bar{x}}$$

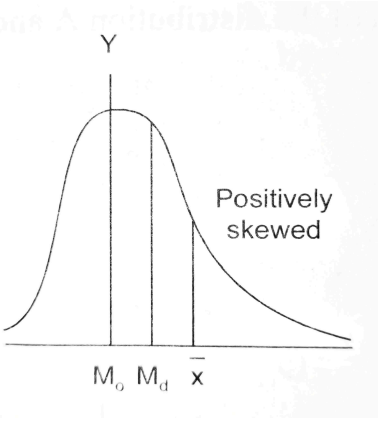
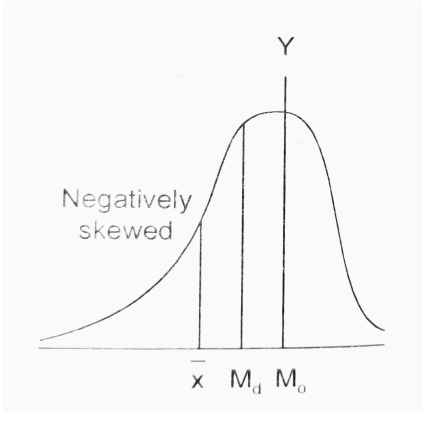
The coefficient of dispersion based on standard deviation multiplied by 100, is known as the coefficient of variation (C.V.). It is given by

$$\text{C.V.} = \frac{\sigma}{\bar{x}} \times 100$$

It is independent of units. Less the C.V., more will be the uniformity, consistency, etc. and more the C.V., less will be the uniformity, consistency, etc.

#Skewness

A distribution which is not symmetrical is known as skewed. So, skewness is the lack of symmetry. It gives the degree of departure from the symmetry. It relates to the shape of the curve of the frequency distribution rather than its size. With view of elongation of the tail of the curve of the frequency distribution, skewness can be divided into two types:

Positive Skewness	Negative Skewness
Skewness is said to be positive when the tail of the curve of the frequency distribution elongates more on the right. If skewness is positive then $\text{Mean} > \text{Median} > \text{Mode}$. 	Skewness is said to be negative when the tail of the curve of the frequency distribution elongates more on the left. If skewness is negative then $\text{Mean} < \text{Median} < \text{Mode}$. 

Measure of skewness

The absolute measure of skewness are given by the following formula:

1. The first measure of skewness
$$= \text{Mean} - \text{Mode} \quad \text{or,} \quad \text{Mean} - \text{Median}$$
2. The second measure of skewness
$$= (Q_3 - M_d) - (M_d - Q_1)$$
$$= Q_3 + Q_1 - 2M_d$$

#Karl Pearson's Coefficient of Skewness

The Karl Pearson's Coefficient of Skewness is given by

$$S_k(P) = \frac{\bar{x} - M_o}{\sigma}$$

This is also known as Pearsonial coefficient of skewness.

For bi-modal distribution, we use the empirical relation:

$$M_o = 3M_d - 2\bar{x}$$

$$\text{So, } S_k(P) = \frac{3(\bar{x} - M_d)}{\sigma}$$

Note: Karl Pearson's Coefficient of Skewness lies between -3 and +3.

For coefficient of skewness, we have the following conclusion:

1. If $S_k = 0$, the distribution is symmetrical.
2. If $S_k > 0$ then the distribution is positively skewed.
3. If $S_k < 0$ then the distribution is negatively skewed.